

# COMPUTER AIDED DRUG DESIGN

## BIO563

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Department of Computational Biology

TA:

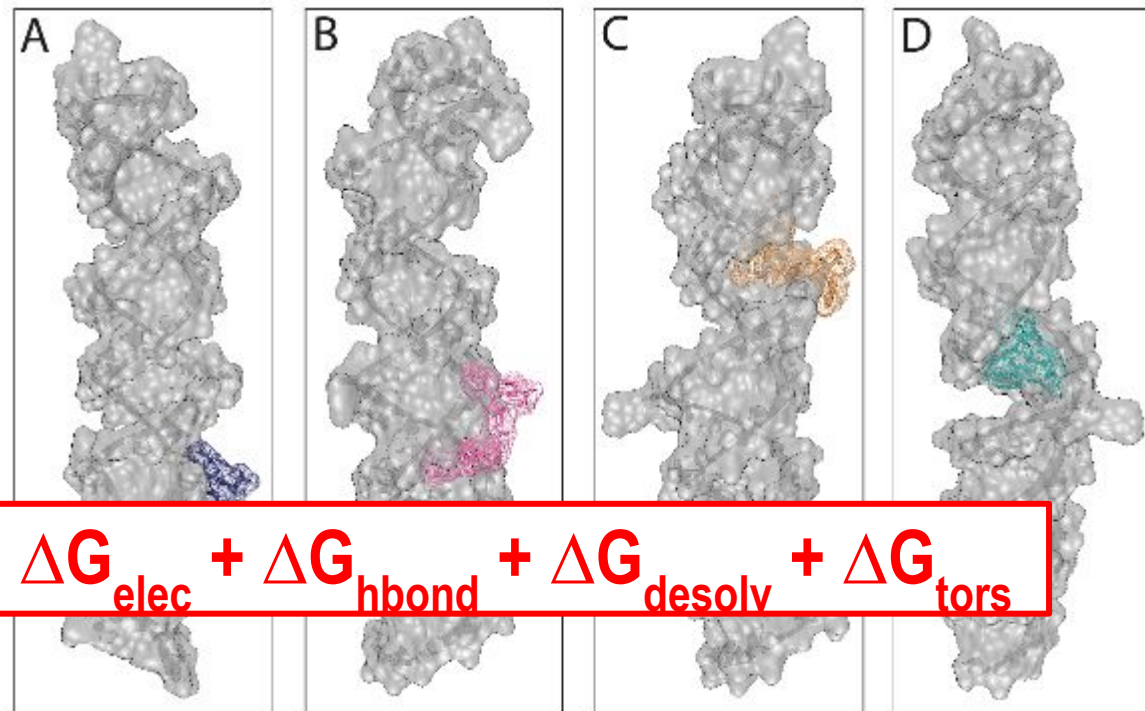
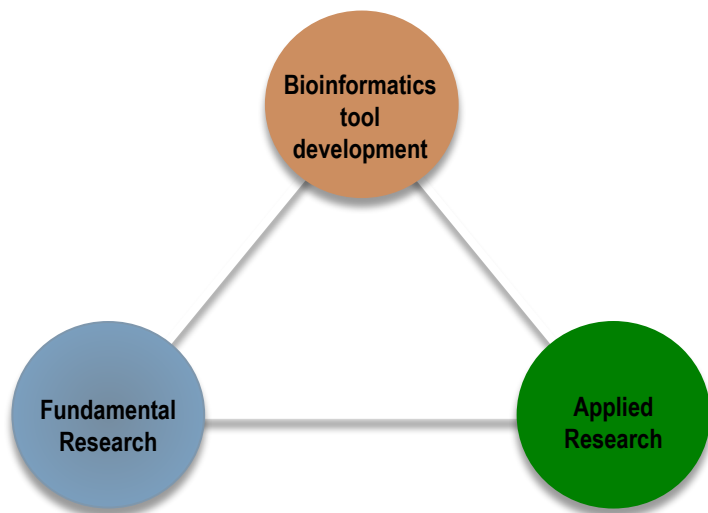
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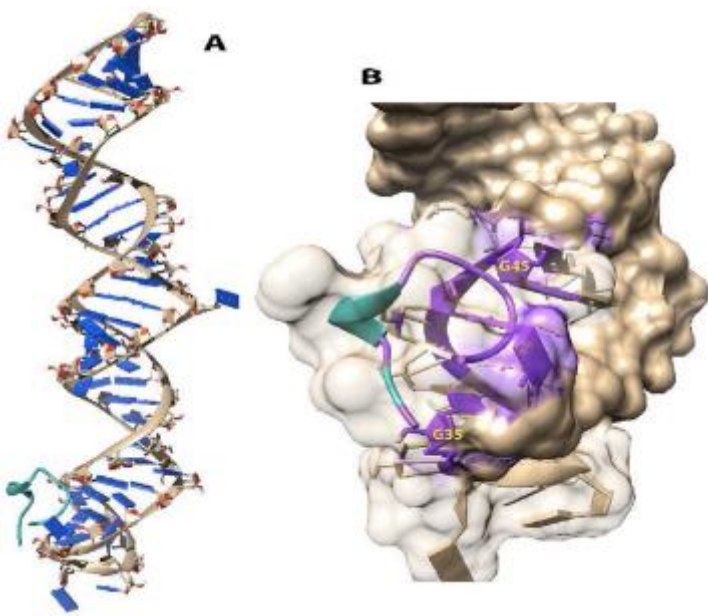
# Applied Research

Ghosh, A., et al.(2018). *MedChemComm*,9(7), 1147-1154.

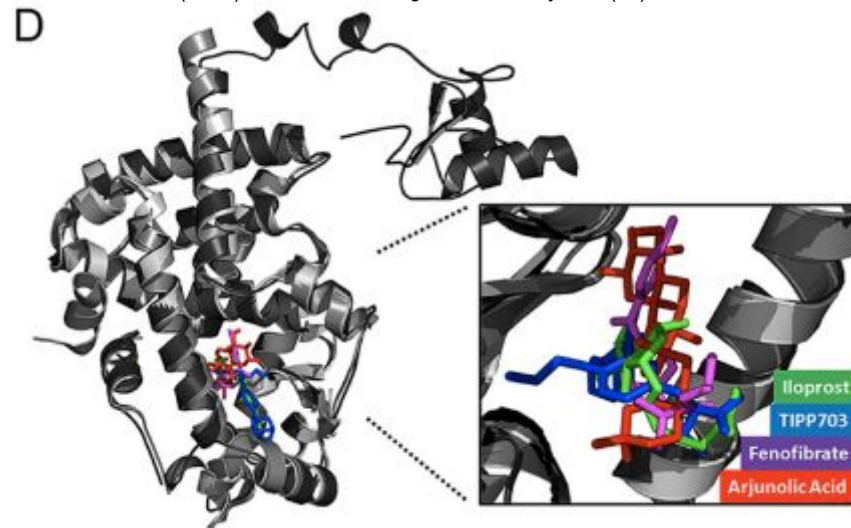


$$\Delta G_{\text{binding}} = \Delta G_{\text{vdW}} + \Delta G_{\text{elec}} + \Delta G_{\text{hbond}} + \Delta G_{\text{desolv}} + \Delta G_{\text{tors}}$$

Bose, D., et al.  
4342-4352.



Bansal, T. et al. (2017). *Journal of Biological Chemistry*, 292(40), 16440-16462.



# COURSE DETAILS

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# COURSE ASSESSMENT

|                        | Share % |
|------------------------|---------|
| Mid-Semester Exam      | 20      |
| End-Semester Exam      | 30      |
| Assignment             | 10      |
| Quizzes (2/3)          | 15      |
| Attendance/Interaction | 10      |
| <b>Group Project</b>   | 15      |

Minimum marks to pass the course: 35

# ATTENDANCE

- $100-80 \% = 10$
- $79-60 = 9$
- $59-40 = 8$
- $39-20 = 7$
- $< 20 = 1$

## GRADING SCHEME (RELATIVE MARKING)

| Grade    | Marks %       |
|----------|---------------|
| <b>F</b> | <b>&lt;35</b> |

Minimum marks NEEDED to be scored to be  
graded in relative scale!

# RESOURCES

- **Molecular Modeling: Principles and Applications**

Andrew R. Leach

- **DRUG DISCOVERY AND DEVELOPMENT – PRESENT AND FUTURE**

Edited by Izet M. Kapetanovic

- **Textbook of Drug Design and Discovery**

*Povl Krosgaard-Larsen, Tommy Liljefors and Ulf Madsen*



# WHAT IS TO BE GAINED FROM THIS COURSE?

- Getting a grasp with basics of structure-ligand relationship
- Thinking on multi-scale approach
- Macro to micro
- Analyzing biological process in spatio-temporal manner
- Realizing the potential of drug discovery!

Preparing for the jobs ahead!



# COURSE EXPECTATIONS

- Self-study of topics and concepts covered in the class  
(Queries/questions are always welcome)
- Group interaction
- Interaction in class is a must!

# POLICY ON PLAGIARISM

“The policy is applicable to all the components of course assessment including assignments, projects, quiz, examinations, and the like. Since the policy is applied across courses, it is essential to report any instance of plagiarism, use of unfair means, **academic dishonesty by students**, or facilitation of any such cases by TAs or TFs, whenever and wherever they may occur, to the Academic Section.”

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|    | Misconduct/use of unfair means in assignments/projects                               | Penalty                                                                                                                             |
|----|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1. | First misconduct/use of unfair means during the entire stay in IIIT-Delhi            | Zero in the assignment (awarded by faculty) + one letter grade less in the course.                                                  |
| 2. | Second misconduct/use of unfair means during entire stay in IIIT-Delhi               | Student is assigned an F grade in the course.                                                                                       |
| 3. | Third or further misconduct/use of unfair means during the entire stay in IIIT-Delhi | Student is assigned an F grade in the course and the case is reported to DAC, who may suspend the student for 1 semester to a year. |

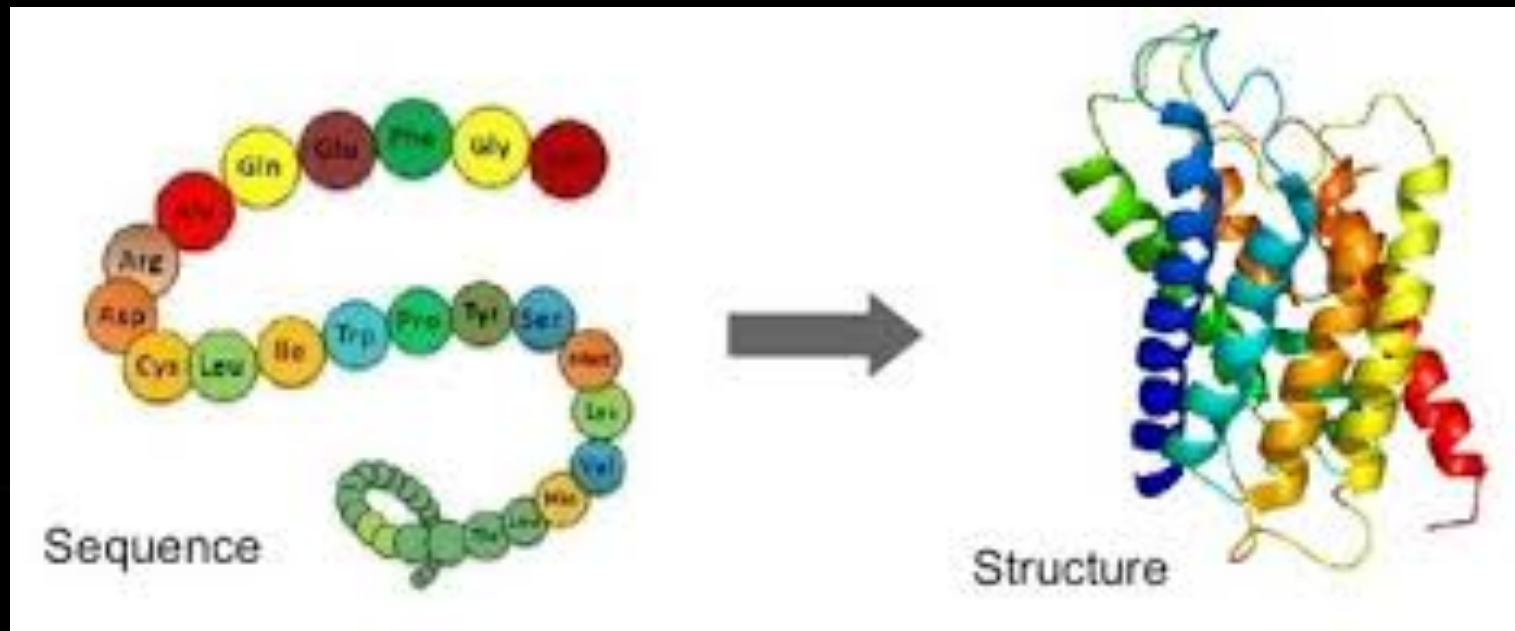
|    | Misconduct/use of unfair means in quiz/midsem/endsem                       | Penalty                                                                                                              |
|----|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 1. | First misconduct/use of unfair means during the entire stay in IIIT-Delhi  | Student is assigned an F grade in the course.                                                                        |
| 2. | Second misconduct/use of unfair means during the entire stay in IIIT-Delhi | Student is assigned an F grade in the course and student may be suspended from the program for 1-2 semesters by DAC. |
| 3. | Third misconduct/use of unfair means during the entire stay in IIIT-Delhi  | Student is assigned an F grade in the course and student's program may be terminated by DAC.                         |

# DIVISION OF GROUPS

- Group will have the responsibility of
  - Project
  - Group assignments
- The group members are going to be assessed on cumulative grading. In case of a non-functional member, it is the responsibility of the group to bring it to the notice of the TA/Instructor.
- It shall be the group's responsibility to ensure the submitted assignment/project is in tune with the policy of plagiarism.

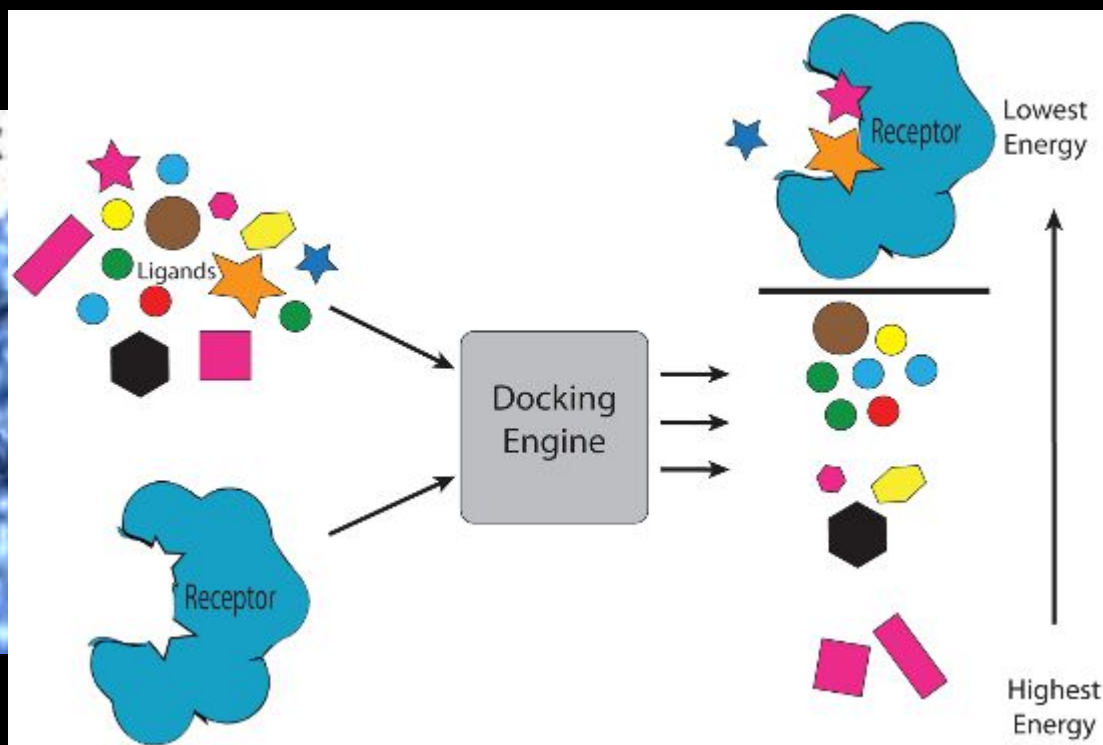
# TECHNIQUES THAT WILL BE TAUGHT

- Structure Prediction: 1D to 3D
- Molecular dynamics (basics)
- Virtual Screening
- Pharmacophore modeling

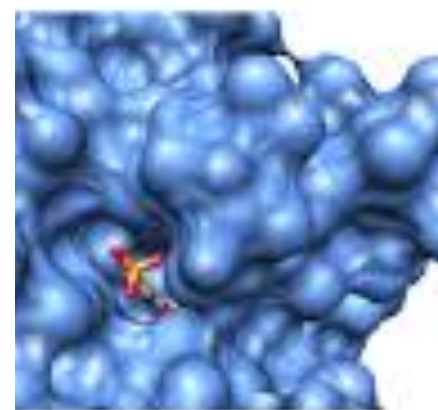


# TECHNIQUES THAT WILL BE TAUGHT

- Structure Prediction: 1D to 3D
- Molecular dynamics (basics)
- Molecular docking
- Virtual screening of drug molecules

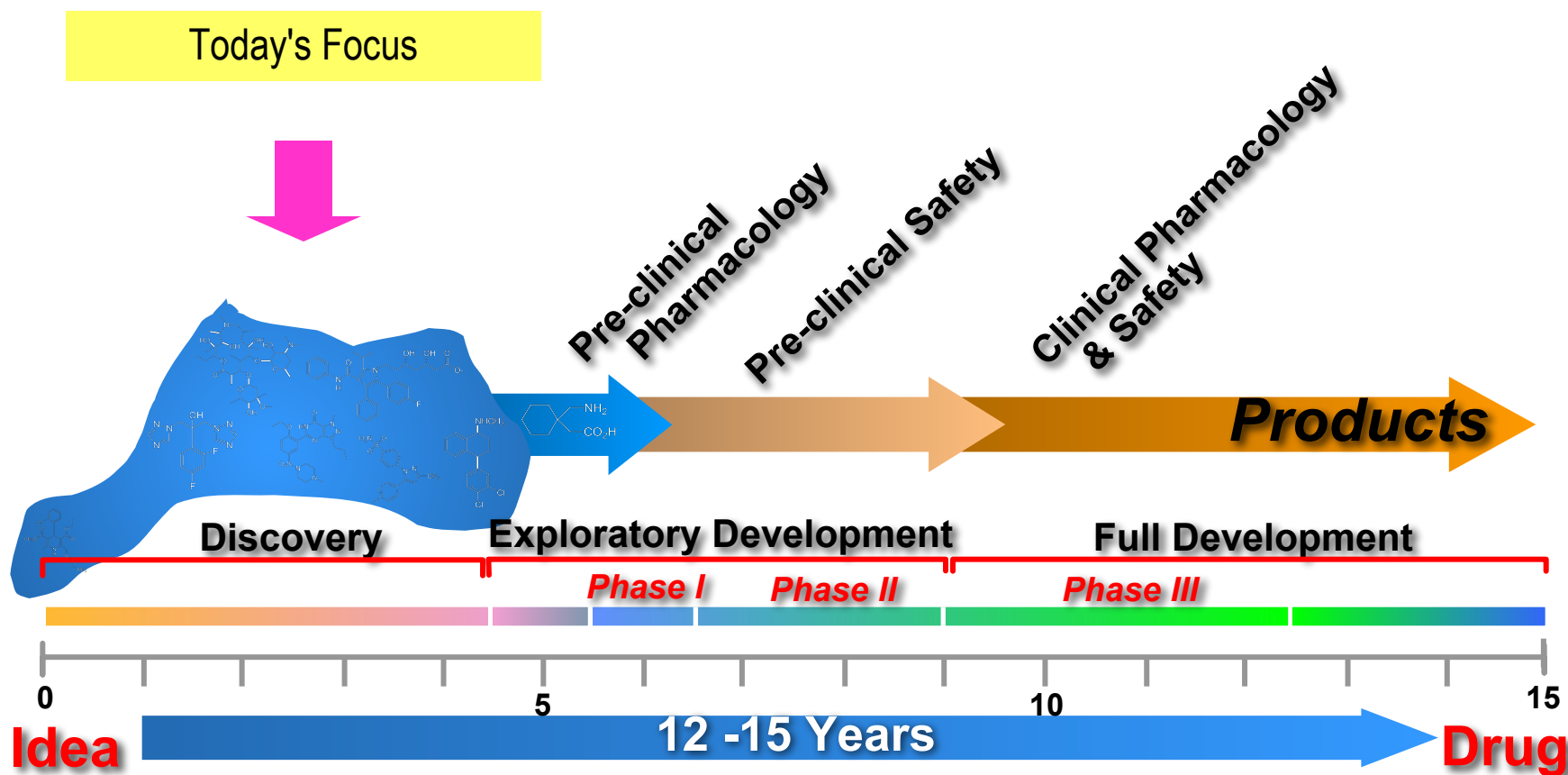


Molecular Docking



# Drug Discovery/Development Pipeline

- Multifaceted, complicated, lengthy process



- Iterative process impacted by technology

